INDIAN INSTITUTE OF TECHNOLOGY (ISM) DHANBAD

DEPARTMENT OF ENVIRONMENTAL SCIENCE AND ENGINEERING

Mid-Semester Examination (Monsoon Semester: 2025-2026)

Environmental Sciences (NESV101)

Date: 20.09.2025 **Duration:** 1 hour

Time: 6:00 PM to 7:00 PM **Total Marks:** 30

Note

All questions are compulsory

Please answer the questions in order

Unit 1

Question 1a: Mention the 4R's of Waste Management [2 Marks]

Question 1b: A solid waste sample with the empirical formula $C_{50}H_{80}O_{40}N_2$ is to be treated in an in-vessel forced-aeration composting system. The waste has a biodegradable volatile solid content of 60% that is completely oxidized during composting. The stochiometric oxidation is given by:

$$\frac{4a + b - 2c - 3d}{4}$$

Where a, b, c, and d are the moles of C, H, O, and N respectively. Determine the amount of air (in m^3) required to compost 1 tonne of waste. Assume that air contains 23% oxygen by weight, has a density of 1.29 kg/m^3, and apply a safety factor of 1.5 to account for design margin. 3492.4 [5.5 Marks]

Unit 2

Question 2a: An adult inhales 12 m^3 air/day. If indoor CO varies as:

- Morning (6 hrs): 10 ppm
- Afternoon (8 hrs): 15 ppm
- Evening (10 hrs): 20 ppm

Calculate total daily CO intake (mg/day) at 25°C, 1 atm (MW CO = 28 g/mol; molar volume = 24.45 L/mol) [3 Marks]

Question 2b: Differentiate between Lofting and Fumigating Plumes on the basis of (i) Atmospheric Condition, (ii) Pollutant Dispersion Pattern, and (iii) Ground-Level Impact. [4.5 Marks]

Unit 3

Question 3a: Determine the fraction of mass of the atmosphere in both the troposphere and stratosphere, respectively. Assume the density of the atmosphere at surface as 1.25 kg/m^3. [5 marks]

P.T.O

$2a + \frac{b}{2}$

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Question 3b: How does the temperature vary with the altitude – vertical structure of temperature – in the earth's atmosphere? Explain – with a proper reasoning – as to why the different layer, with accord to temperature variation that exists in our atmosphere. [2.5 marks]

Unit 4

Question 4a: Determine the 1-day BOD and ultimate BOD for a wastewater whose 5-day 20^{0}C BOD is 200 mg/L. The reaction constant k (base e) = 0.23 d^{-1}. What would have been the 5-day BOD if the test had been conducted at 25^{0}C? [5 marks]

23.964, 300.42, 0.2893

Question 4b: Give a schematic representation of a typical surface water treatment plant. [2.5 marks]

Aeration → Screening → Flocculation → Sedimentation

↓ Filtration

Disinfectant contact ↓ Clearwell reservoir

↓ distri

1.15